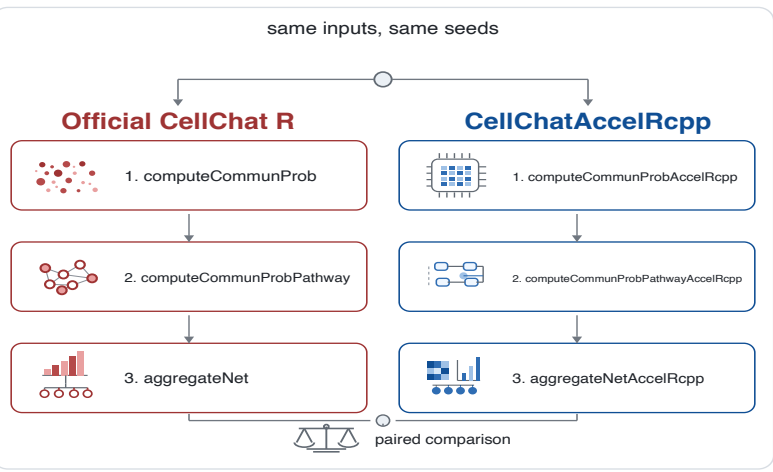


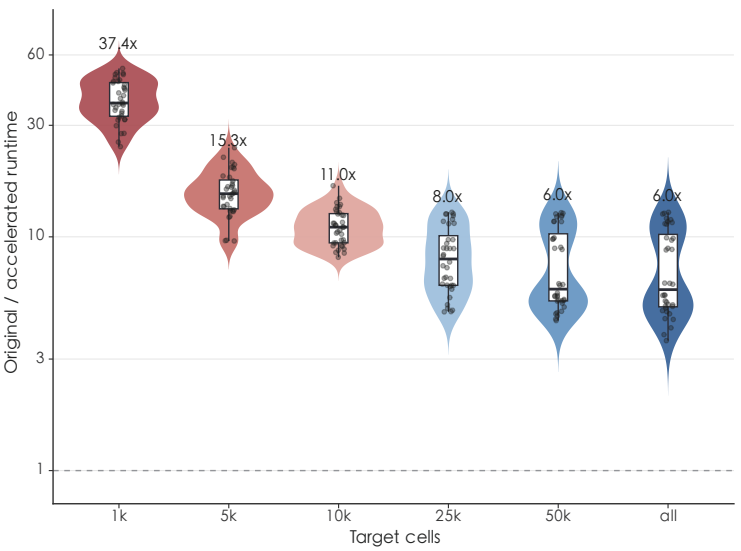
A

Paired-engine comparison



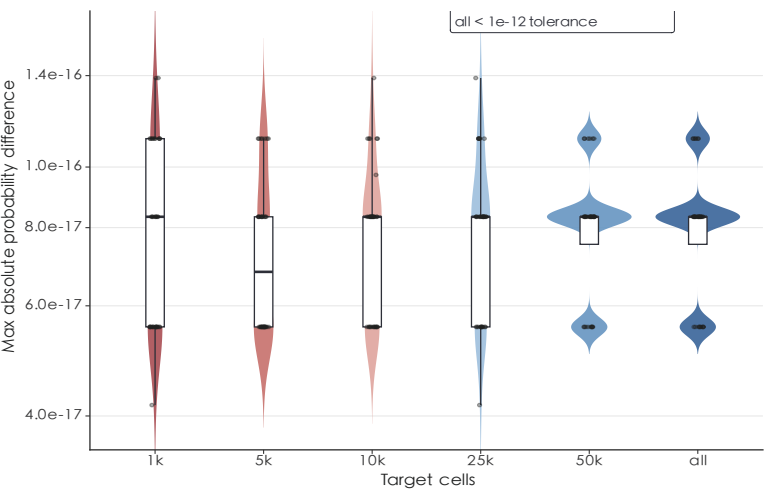
C

Speedup distribution



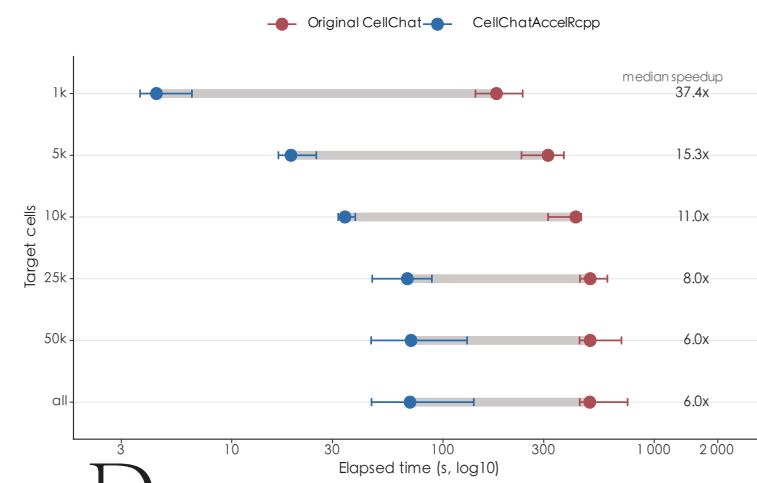
E

Numerical equivalence to original CellChat



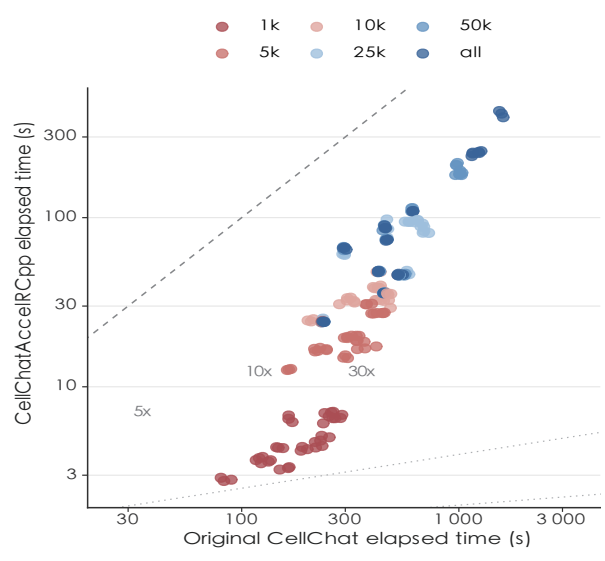
B

Runtime compression across cell scales



D

Paired runtime comparison



F

Component ablation identifies the dominant acceleration target

